

2016.0 RANGE ROVER (LG), 412-00

# CLIMATE CONTROL SYSTEM – GENERAL INFORMATION

DIAGNOSIS AND TESTING

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## PRINCIPLES OF OPERATION

For a detailed description of the Air Conditioning system, refer to the relevant Description and Operation section in the workshop manual.

REFER to: [Air Conditioning](#) (412-01B Climate Control - SDV6 3.0L Diesel - Hybrid Electric Vehicle, Description and Operation).

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## INSPECTION AND VERIFICATION

### WARNINGS:

- A risk assessment must be performed before any work is undertaken.
- All hybrid electric vehicle work must be performed by a suitably qualified person.
- Appropriate Personal Protection Equipment (PPE) must be worn when working on or near a hybrid electric vehicle high voltage system.
- The hybrid electric vehicle battery pack casing must not be opened due to the risk of exposure to hazardous voltage, which may cause serious injury or death.

- The electric power inverter converter casing must not be disassembled due to the risk of exposure to hazardous voltage, which may cause serious injury or death.
- If the hybrid electric vehicle battery pack is damaged or overcharged, there is a risk of exposure to hazardous voltage and/or highly corrosive electrolyte mist. If liquid or vapour is observed leaking from the hybrid electric vehicle battery pack, take the following action:
  - Evacuate the area
  - Notify manager
  - Do not breathe smoke/vapour
  - Contain spillage using spill kit
  - Wash any spillage off body and clothing (remove contaminated clothing)
- If the hybrid electric vehicle warning indicator is illuminated, the Battery Energy Control Module (BECM) cannot isolate the high voltage cables. Live working will be necessary to rectify the fault.
- The motor generator can generate a voltage if the wheels are rotated, even if the hybrid electric vehicle system has been made safe.
- Servicing must be carried out by personnel familiar with both vehicle system and the charging and testing equipment. All operations must be carried out in a well ventilated area away from open flame and heat sources.

## CAUTIONS:

- Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.
- When probing connectors to take measurements in the course of the pinpoint tests, use the adaptor kit, part number 3548-1358-00.

## NOTES:

- If the control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual (section B1.2), or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system).
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance, always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required.

**1.** Verify the customer concern

**2.** Visually inspect for obvious signs of damage and system integrity

## Visual Inspection

| MECHANICAL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | ELECTRICAL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>▪ Coolant level</li> <li>▪ Hose(s)</li> <li>▪ Coolant pump</li> <li>▪ Control flap(s)</li> <li>▪ Duct(s)</li> <li>▪ Vent(s)</li> <li>▪ Cabin air filter</li> <li>▪ Electric air conditioning compressor</li> <li>▪ Thermostatic expansion valve</li> <li>▪ Evaporator</li> <li>▪ Receiver drier</li> <li>▪ Air conditioning condenser</li> <li>▪ Refrigerant pipes</li> <li>▪ Auxiliary drive belt</li> <li>▪ Fuel fired booster heater</li> <li>▪ Fuel fired booster heater fuel pump</li> <li>▪ Fuel fired booster heater fuel pipes</li> <li>▪ Hybrid electric vehicle battery pack cooling system</li> <li>▪ Battery pack cooler and isolation valve</li> <li>▪ Battery pack cooler and and thermal expansion valve</li> </ul> | <ul style="list-style-type: none"> <li>▪ Fuse(s)</li> <li>▪ Wiring harnesses and connectors</li> <li>▪ High voltage cables</li> <li>▪ Automatic Temperature Control Module (ATCM)</li> <li>▪ Battery Energy Control Module (BECM)</li> <li>▪ Electric Power Inverter Converter Module (EPICM)</li> <li>▪ Hybrid electric vehicle battery pack</li> <li>▪ Refrigerant pressure sensor</li> <li>▪ Blower</li> <li>▪ Electric cooling fan</li> <li>▪ Electric air conditioning compressor</li> <li>▪ Battery energy control module electric coolant pump</li> </ul> |

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step
4. If the cause is not visually evident, verify the symptom and refer to the Symptom Chart, alternatively check for Diagnostic Trouble Codes (DTCs) and refer to the DTC Index
5. Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required

## SYMPTOM CHART

| SYMPTOM                                                 | POSSIBLE CAUSES                                                                                                                                                                                                               | ACTION                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Air conditioning performance poor or inoperative        | <ul style="list-style-type: none"><li>▪ Refrigerant undercharged</li><li>▪ Refrigerant overcharged</li><li>▪ Thermostatic expansion valve faulty</li><li>▪ Receiver drier restricted</li><li>▪ Water in refrigerant</li></ul> | <ul style="list-style-type: none"><li>▪ GO to Pinpoint Test <a href="#">A.</a></li></ul>                                                                                                                                                                                                                                                                               |
| Air conditioning operates briefly and then switches off | <ul style="list-style-type: none"><li>▪ Electric cooling fan inoperative</li><li>▪ Air conditioning condenser airflow obstructed</li></ul>                                                                                    | <ul style="list-style-type: none"><li>▪ Test the operation of the electric cooling fan</li><li>▪ Check the air conditioning condenser for external obstructions</li></ul>                                                                                                                                                                                              |
| Air conditioning inoperative                            | <ul style="list-style-type: none"><li>▪ Battery energy control module fault</li><li>▪ Electric power inverter converter module fault</li></ul>                                                                                | <ul style="list-style-type: none"><li>▪ Using the manufacturer approved diagnostic system, check the battery energy control module for related DTCs and refer to the relevant DTC index</li><li>▪ Using the manufacturer approved diagnostic system, check the electric power inverter converter module for related DTCs and refer to the relevant DTC index</li></ul> |

## STEPPER MOTORS

| SYMPTOM     | POSSIBLE CAUSES | ACTION    |
|-------------|-----------------|-----------|
| LACK OF AIR | MISSING         | USING THE |

BEING  
EMITTED  
THROUGH  
THE VARIOUS  
AIR VENTS  
WHEN  
SELECTED

INITIALIZATION/CALIBRATION  
FOR STEPPER MOTOR(S)

MANUFACTURER  
APPROVED  
DIAGNOSTIC SYSTEM,  
CHECK THE AUTOMATIC  
TEMPERATURE  
CONTROL MODULE FOR  
RELATED DTCS AND  
REFER TO THE  
RELEVANT DTC INDEX

IF STEPPER MOTOR  
RELATED DTCS LOGGED  
OTHER THAN THOSE  
LISTING IN STEPPER  
MOTORS SYMPTOM  
SECTION THEN  
INVESTIGATE AND  
RECTIFY, OTHERWISE  
FOLLOW STEP BELOW

USING THE  
MANUFACTURER  
APPROVED  
DIAGNOSTIC SYSTEM  
CLEAR THE DTCS THEN  
PERFORM ROUTINE -  
ACTUATOR  
INITIALIZATION  
(0X900D). CARRY OUT  
THE STEPPER MOTOR  
INITIALIZATION  
PROCESS AND RETEST.  
PERFORM ROUTINE –  
ON DEMAND SELF TEST  
(0X0202). CHECK THE  
AUTOMATIC  
TEMPERATURE  
CONTROL MODULE FOR  
RELATED DTCS AND  
REFER TO THE  
RELEVANT DTC INDEX

IF AFTER FOLLOWING  
STEPPER MOTOR  
INITIALIZATION  
PROCESS THERE ARE  
NO FURTHER DTCS AND  
AIR FLOW IS STILL  
LACKING IN ONE OR  
MORE VENTS THEN THE  
FAULT IS NOT  
ELECTRICAL.

INVESTIGATE FOR  
BROKEN LINKAGES OR  
BLOCKAGE IN DUCTS

- |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                              |                                                                                            |
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| <ul style="list-style-type: none"> <li>▪ Recirculation motor not engaging/stuck in a position, DTCs - B1083-49, B1083-87, B1083-97</li> <li>▪ Left front temperature blend motor not engaging/stuck in a position, DTCs - B1352-49, B1352-87, B1352-97</li> <li>▪ Right front temperature blend motor not engaging/stuck in a position, DTCs - B1366-49, B1366-87, B1366-97</li> <li>▪ Left rear temperature blend motor not engaging/stuck in a position, DTCs - B1364-49, B1364-87, B1364-97</li> <li>▪ Right rear temperature blend motor not engaging/stuck in a position, DTCs - B1354-49, B1354-87, B1354-97</li> <li>▪ Rear temperature blend motor not</li> </ul> | <ul style="list-style-type: none"> <li>▪ Stepper motor jammed or broken</li> <li>▪ Wiring fault</li> <li>▪ Stepper motor internal failure</li> <li>▪ Automatic temperature control module failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ GO to Pinpoint Test <a href="#">E</a>.</li> </ul> |
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engaging/stuck  
in a position,  
DTCs - B1435-  
49, B1435-87,  
B1435-97

- Left auxiliary  
climate control  
temperature  
blend motor  
not  
engaging/stuck  
in a position,  
DTCs - B13F2-  
49, B13F2-87,  
B13F2-97

- Right auxiliary  
climate control  
temperature  
blend motor  
not  
engaging/stuck  
in a position,  
DTCs - B13F3-  
49, B13F3-87,  
B13F3-97

- Front cool air  
bypass motor  
not  
engaging/stuck  
in a position,  
DTCs - B13EE-  
49, B13EE-87,  
B13EE-97

- Demist  
distribution  
motor not  
engaging/stuck  
in a position,  
DTCs - B1085-  
49, B1085-87,  
B1085-97

- Left front  
face/feet  
distribution  
motor not  
engaging/stuck  
in a position,  
DTCs - B13EF-  
49, B13EF-87,  
B13EF-97)

- Right front

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
| <p>face/feet<br/>distribution<br/>motor not<br/>engaging/stuck<br/>in a position,<br/>DTCs - B13F0-<br/>49, B13F0-87,<br/>B13F0-97)</p> <ul style="list-style-type: none"> <li>▪ Rear face/feet<br/>distribution<br/>motor not<br/>engaging/stuck<br/>in a position,<br/>DTCs - B1B7D-<br/>49, B1B7D-87,<br/>B1B7D-97)</li> <li>▪ Front face/feet<br/>distribution<br/>motor not<br/>engaging/stuck<br/>in a position,<br/>DTCs - B1434-<br/>49, B1434-87,<br/>B1434-97)</li> <li>▪ Auxiliary<br/>climate control<br/>face/feet<br/>distribution<br/>motor not<br/>engaging/stuck<br/>in a position,<br/>DTCs - B13F1-<br/>49, B13F1-87,<br/>B13F1-97)</li> </ul> |  |  |
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## PINPOINT TESTS

### NOTE:

This test is performed with the engine **not** running.

### PINPOINT TEST A : PRELIMINARY TESTS

| TEST<br>CONDITIONS     | DETAILS/RESULTS/ACTIONS                                                                                                                                                  |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A1: PRELIMINARY TEST 1 |                                                                                                                                                                          |
|                        | 1 Close the valves on the air conditioning station                                                                                                                       |
|                        | 2 Connect the air conditioning station to the vehicle charging ports                                                                                                     |
|                        | 3 Record the pressure values                                                                                                                                             |
|                        | <p>Is a pressure registered on both gauges?</p> <p><b>Yes</b><br/>GO to Pinpoint Test <a href="#">B.</a></p> <p><b>No</b><br/>GO to Pinpoint Test <a href="#">D.</a></p> |

| PINPOINT TEST B : FUNCTIONALITY TESTS |                                                                                                                                                                                                                                                            |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TEST<br>CONDITIONS                    | DETAILS/RESULTS/ACTIONS                                                                                                                                                                                                                                    |
| B1: FUNCTIONALITY TEST 1              |                                                                                                                                                                                                                                                            |
|                                       | 1 Close the valves on the air conditioning station                                                                                                                                                                                                         |
|                                       | 2 Connect the air conditioning station to the vehicle charging ports                                                                                                                                                                                       |
|                                       | 3 Open all doors and the tailgate                                                                                                                                                                                                                          |
|                                       | 4 Start the engine                                                                                                                                                                                                                                         |
|                                       | 5 Set the temperature to the lowest setting (all zones)                                                                                                                                                                                                    |
|                                       | 6 Set the fan speed to maximum                                                                                                                                                                                                                             |
|                                       | 7 Set the recirculate switch to off                                                                                                                                                                                                                        |
|                                       | 8 Insert a temperature probe into the centre air vent                                                                                                                                                                                                      |
|                                       | 9 Raise engine speed to 1500rpm and maintain this speed for 5 minutes                                                                                                                                                                                      |
|                                       | 10 Record the temperature value                                                                                                                                                                                                                            |
|                                       | 11 Record the pressure values                                                                                                                                                                                                                              |
|                                       | <p>Are the pressure readings stable and in the green 'normal' region of the gauge?</p> <p><b>Yes</b><br/>Air conditioning system operating normally</p> <p><b>No</b><br/>Air conditioning system fault present. GO to Pinpoint Test <a href="#">C.</a></p> |

**NOTE:**

This test is performed with the engine running and the air conditioning set to on.

| PINPOINT TEST C : GAUGE TESTS |                                                                                                                                                                                                                                                                                                                                                   |
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| TEST CONDITIONS               | DETAILS/RESULTS/ACTIONS                                                                                                                                                                                                                                                                                                                           |
| C1: GAUGE TEST 1              |                                                                                                                                                                                                                                                                                                                                                   |
|                               | 1 Record the pressure values                                                                                                                                                                                                                                                                                                                      |
|                               | Did the gauges register a change in pressure when the air conditioning was switched on?<br><b>Yes</b><br><b>GO to C2.</b><br><b>No</b><br>Using the manufacturer approved diagnostic system, check the Automatic Temperature Control Module (ATCM) for related DTCs and refer to the relevant DTC index                                           |
| C2: GAUGE TEST 2              |                                                                                                                                                                                                                                                                                                                                                   |
|                               | 1 Record the pressure values                                                                                                                                                                                                                                                                                                                      |
|                               | Are the pressure gauge readings fluctuating?<br><b>Yes</b><br>Moisture present in the air conditioning system. Recover the refrigerant. Install a new receiver drier. Refer to the relevant section of the workshop manual and evacuate and recharge the air conditioning system. GO to Pinpoint Test <b>B</b> .<br><b>No</b><br><b>GO to C3.</b> |
| C3: GAUGE TEST 3              |                                                                                                                                                                                                                                                                                                                                                   |
|                               | 1 Record the pressure values                                                                                                                                                                                                                                                                                                                      |
|                               | Are the pressure gauge readings too low?<br><b>Yes</b><br><b>GO to C4.</b><br><b>No</b><br><b>GO to C6.</b>                                                                                                                                                                                                                                       |
| C4: GAUGE TEST 4              |                                                                                                                                                                                                                                                                                                                                                   |
|                               | 1 Using the manufacturer approved refrigerant leak detector, check for a refrigerant leak                                                                                                                                                                                                                                                         |
|                               |                                                                                                                                                                                                                                                                                                                                                   |

|  |                                                                                                                                                                                                                                                                                                   |
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|  | <p>Was a refrigerant leak detected?</p> <p><b>Yes</b></p> <p>Refer to the relevant section of the workshop manual and recover the refrigerant. Repair as necessary. Evacuate and recharge the air conditioning system. GO to Pinpoint Test <b>B</b>.</p> <p><b>No</b></p> <p><b>GO to C5.</b></p> |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### C5: GAUGE TEST 5

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p><b>1</b> Refer to the relevant section of the workshop manual and recover the refrigerant</p>                                                                                                                                                                                                                                                                                                                                                                           |
|  | <p>Was the weight of the recovered refrigerant less than specified for the air conditioning system?</p> <p><b>Yes</b></p> <p>Refer to the relevant section of the workshop manual and evacuate and recharge the air conditioning system. GO to Pinpoint Test <b>B</b>.</p> <p><b>No</b></p> <p>Install a new receiver drier. Refer to the relevant section of the workshop manual and evacuate and recharge the air conditioning system. GO to Pinpoint Test <b>B</b>.</p> |

#### C6: GAUGE TEST 6

|  |                                                                                                                                                                     |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p><b>1</b> Record the pressure values</p>                                                                                                                          |
|  | <p>Are the pressure gauge readings too high?</p> <p><b>Yes</b></p> <p><b>GO to C7.</b></p> <p><b>No</b></p> <p>Test inconclusive. GO to Pinpoint Test <b>B</b>.</p> |

#### C7: GAUGE TEST 7

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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|  | <p><b>1</b> Refer to the relevant section of the workshop manual and recover the refrigerant</p>                                                                                                                                                                                                                                                                                                                                                                                    |
|  | <p>Was the weight of the recovered refrigerant more than specified for the air conditioning system?</p> <p><b>Yes</b></p> <p>Refer to the relevant section of the workshop manual and evacuate and recharge the air conditioning system. GO to Pinpoint Test <b>B</b>.</p> <p><b>No</b></p> <p>Install a new thermal expansion valve. Refer to the relevant section of the workshop manual and evacuate and recharge the air conditioning system. GO to Pinpoint Test <b>B</b>.</p> |

## CAUTION:

When charging the system with nitrogen, the pressure should be regulated to 7 bar.

## NOTE:

This test is performed with the engine **not** running.

| PINPOINT TEST D : NITROGEN LEAK TESTS |                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TEST CONDITIONS                       | DETAILS/RESULTS/ACTIONS                                                                                                                                                                                                                                                                                                                                                                                                         |
| D1: NITROGEN LEAK TEST                |                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                       | 1 Charge the air conditioning system with nitrogen                                                                                                                                                                                                                                                                                                                                                                              |
|                                       | 2 Isolate the nitrogen supply                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                       | 3 Monitor the pressure gauge and check for leaks                                                                                                                                                                                                                                                                                                                                                                                |
|                                       | Has the source of the leak been identified?<br><b>Yes</b><br>Rectify the leak as necessary. Install a new receiver drier. Refer to the relevant section of the workshop manual and evacuate and recharge the air conditioning system. GO to Pinpoint Test <b>B</b> .<br><b>No</b><br>Refer to the relevant section of the workshop manual and evacuate and recharge the air conditioning system. GO to Pinpoint Test <b>B</b> . |

| PINPOINT TEST E : STEPPER MOTORS                                 |                                                                     |
|------------------------------------------------------------------|---------------------------------------------------------------------|
| TEST CONDITIONS                                                  | DETAILS/RESULTS/ACTIONS                                             |
| E1: RECIRCULATION MOTOR OPERATION (B1083-49, B1083-87, B1083-97) |                                                                     |
|                                                                  | 1 Using the manufacturer approved diagnostic system, clear the DTCs |
|                                                                  | 2 Remove glove compartment so recirculation door is visible         |
|                                                                  | 3 Turn ignition on and A/C on, set temperature to 28/28             |
|                                                                  | 4 Toggle Recirculation button on and off                            |
|                                                                  |                                                                     |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>Does recirculation door move between fresh &amp; recirculate positions?</p> <p><b>Yes</b></p> <p>Check the DTC's - has the DTC recurred? <b>A)</b> If the DTC has recurred go back to step 1 and repeat - if the DTC consistently recurring, check for obstruction/mechanical linkage problem/damage to recirculation door, if fault found repair. Clear the DTCs and retest. <b>B)</b> If the DTC has not recurred - no fault found</p> <p><b>No</b></p> <p>Check for obstruction/mechanical linkage problem/damage to recirculation door, if fault found repair. Clear the DTCs and retest. Check wiring - does recirculation motor have power, ground and LIN connection to ECU? <b>A)</b> If wiring or LIN fault found - repair wiring. Clear the DTCs and retest. <b>B)</b> If no fault found, and there is sufficient power, ground and LIN connection is good, install a new recirculation motor. Clear the DTCs and retest</p> |
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## E2: LEFT FRONT TEMPERATURE BLEND MOTOR OPERATION (B1352-49, B1352-87, B1352-97)

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | 1 Using the manufacturer approved diagnostic system, clear the DTCs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|  | 2 Start the engine                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|  | 3 Turn A/C on and set to Hi/Hi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|  | 4 Run engine to operating temperature to allow ATCM to provide hot air                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|  | 5 Toggle A/C settings from LO/LO to HI/Hi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|  | <p>When in LO/LO is air from left front face vent cold (&lt;10'c) and when in HI/Hi is air from left front face vent hot (&gt;50'c)?</p> <p><b>Yes</b></p> <p>Check the DTC's - has the DTC recurred? <b>A)</b> If the DTC has recurred go back to step 1 and repeat - if the DTC consistently recurring, check for obstruction/mechanical linkage problem/damage to recirculation door, if fault found repair. Clear the DTCs and retest. <b>B)</b> If the DTC has not recurred - no fault found</p> <p><b>No</b></p> <p>Check for obstruction/mechanical linkage problem/damage to recirculation door, if fault found repair. Clear the DTCs and retest. Check wiring - does left front temperature blend motor have power, ground and LIN connection to ECU? <b>A)</b> If wiring or LIN fault found - repair wiring. Clear the DTCs and retest. <b>B)</b> If no fault found, and there is sufficient power, ground and LIN connection is good, install a new left front temperature blend motor. Clear the DTCs and retest</p> |

## E3: RIGHT FRONT TEMPERATURE BLEND MOTOR OPERATION (B1366-49, B1366-87, B1366-97)

|  |                                                                     |
|--|---------------------------------------------------------------------|
|  | 1 Using the manufacturer approved diagnostic system, clear the DTCs |
|  | 2 Start the engine                                                  |
|  | 3 Turn A/C on and set to Hi/Hi                                      |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>4</b> Run engine to operating temperature to allow ATCM to provide hot air                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|  | <b>5</b> Toggle A/C settings from LO/LO to HI/HI                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|  | <p>When in LO/LO is air from right front face vent cold (&lt;10°C) and when in HI/HI is air from right front face vent hot (&gt;50°C)?</p> <p><b>Yes</b></p> <p>Check the DTC's - has the DTC recurred? <b>A)</b> If the DTC has recurred go back to step 1 and repeat - if the DTC consistently recurring, check for obstruction/mechanical linkage problem/damage to recirculation door, if fault found repair. Clear the DTCs and retest. <b>B)</b> If the DTC has not recurred - no fault found</p> <p><b>No</b></p> <p>Check for obstruction/mechanical linkage problem/damage to recirculation door, if fault found repair. Clear the DTCs and retest. Check wiring - does right front temperature blend motor have power, ground and LIN connection to ECU? <b>A)</b> If wiring or LIN fault found - repair wiring. Clear the DTCs and retest. <b>B)</b> If no fault found, and there is sufficient power, ground and LIN connection is good, install a new right front temperature blend motor. Clear the DTCs and retest</p> |

#### E4: LEFT REAR TEMPERATURE BLEND MOTOR OPERATION (B1364-49, B1364-87, B1364-97)

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>1</b> Using the manufacturer approved diagnostic system, clear the DTCs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|  | <b>2</b> Start the engine                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|  | <b>3</b> Turn rear A/C on and set to Hi/Hi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|  | <b>4</b> Run engine to operating temperature to allow ATCM to provide hot air                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|  | <b>5</b> Toggle rear A/C settings from LO/LO to HI/HI                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|  | <p>When in LO/LO is air from left rear face vent cold (&lt;10°C) and when in HI/HI is air from left rear face vent hot (&gt;50°C)?</p> <p><b>Yes</b></p> <p>Check the DTC's - has the DTC recurred? <b>A)</b> If the DTC has recurred go back to step 1 and repeat - if the DTC consistently recurring, check for obstruction/mechanical linkage problem/damage to recirculation door, if fault found repair. Clear the DTCs and retest. <b>B)</b> If the DTC has not recurred - no fault found</p> <p><b>No</b></p> <p>Check for obstruction/mechanical linkage problem/damage to recirculation door, if fault found repair. Clear the DTCs and retest. Check wiring - does left rear temperature blend motor have power, ground and LIN connection to ECU? <b>A)</b> If wiring or LIN fault found - repair wiring. Clear the DTCs and retest. <b>B)</b> If no fault found, and there is sufficient power, ground and LIN connection is good, install a new left rear temperature blend motor. Clear the DTCs and retest</p> |

#### E5: RIGHT REAR TEMPERATURE BLEND MOTOR OPERATION (B1354-49, B1354-



|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>1</b> Using the manufacturer approved diagnostic system, clear the DTCs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|  | <b>2</b> Start the engine                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|  | <b>3</b> Turn rear A/C on and set to Hi/Hi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|  | <b>4</b> Run engine to operating temperature to allow ATCM to provide hot air                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|  | <b>5</b> Toggle rear A/C settings from LO/LO to HI/Hi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|  | <p>When in LO/LO is air from right rear face vent cold (&lt;10'c) and when in HI/Hi is air from right rear face vent hot (&gt;50'c)?</p> <p><b>Yes</b></p> <p>Check the DTC's - has the DTC recurred? <b>A)</b> If the DTC has recurred go back to step 1 and repeat - if the DTC consistently recurring, check for obstruction/mechanical linkage problem/damage to recirculation door, if fault found repair. Clear the DTCs and retest. <b>B)</b> If the DTC has not recurred - no fault found</p> <p><b>No</b></p> <p>Check for obstruction/mechanical linkage problem/damage to recirculation door, if fault found repair. Clear the DTCs and retest. Check wiring - does right rear temperature blend motor have power, ground and LIN connection to ECU? <b>A)</b> If wiring or LIN fault found - repair wiring. Clear the DTCs and retest. <b>B)</b> If no fault found, and there is sufficient power, ground and LIN connection is good, install a new right rear temperature blend motor. Clear the DTCs and retest</p> |

#### E6: REAR TEMPERATURE BLEND MOTOR OPERATION (B1435-49, B1435-87, B1435-97)

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>1</b> Using the manufacturer approved diagnostic system, clear the DTCs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|  | <b>2</b> Start the engine                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|  | <b>3</b> Turn rear A/C on and set to Hi/Hi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|  | <b>4</b> Run engine to operating temperature to allow ATCM to provide hot air                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|  | <b>5</b> Toggle rear A/C settings from LO/LO to HI/Hi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|  | <p>When in LO/LO is air from rear face vent cold (&lt;10'c) and when in HI/Hi is air from rear face vent hot (&gt;50'c)?</p> <p><b>Yes</b></p> <p>Check the DTC's - has the DTC recurred? <b>A)</b> If the DTC has recurred go back to step 1 and repeat - if the DTC consistently recurring, check for obstruction/mechanical linkage problem/damage to recirculation door, if fault found repair. Clear the DTCs and retest. <b>B)</b> If the DTC has not recurred - no fault found</p> <p><b>No</b></p> <p>Check for obstruction/mechanical linkage problem/damage to recirculation door, if fault found repair. Clear the DTCs and retest. Check</p> |

wiring - does rear temperature blend motor have power, ground and LIN connection to ECU? **A)** If wiring or LIN fault found - repair wiring. Clear the DTCs and retest. **B)** If no fault found, and there is sufficient power, ground and LIN connection is good, install a new rear temperature blend motor. Clear the DTCs and retest

#### E7: LEFT AUXILIARY CLIMATE CONTROL TEMPERATURE BLEND MOTOR OPERATION (B13F2-49, B13F2-87, B13F2-97)

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | 1 Using the manufacturer approved diagnostic system, clear the DTCs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|  | 2 Start the engine                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|  | 3 Turn A/C on and set to Hi/Hi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|  | 4 Run engine to operating temperature to allow ATCM to provide hot air                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|  | 5 Toggle A/C settings from LO/LO to HI/HI                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|  | <p>When in LO/LO is air from left auxiliary vent cold (&lt;10'c) and when in HI/HI is air from left auxiliary vent hot (&gt;50'c)?</p> <p><b>Yes</b></p> <p>Check the DTC's - has the DTC recurred? <b>A)</b> If the DTC has recurred go back to step 1 and repeat - if the DTC consistently recurring, check for obstruction/mechanical linkage problem/damage to recirculation door, if fault found repair. Clear the DTCs and retest. <b>B)</b> If the DTC has not recurred - no fault found</p> <p><b>No</b></p> <p>Check for obstruction/mechanical linkage problem/damage to recirculation door, if fault found repair. Clear the DTCs and retest. Check wiring - does left auxiliary climate control temperature blend motor have power, ground and LIN connection to ECU? <b>A)</b> If wiring or LIN fault found - repair wiring. Clear the DTCs and retest. <b>B)</b> If no fault found, and there is sufficient power, ground and LIN connection is good, install a new left auxiliary climate control temperature blend motor. Clear the DTCs and retest</p> |

#### E8: RIGHT AUXILIARY CLIMATE CONTROL TEMPERATURE BLEND MOTOR OPERATION (B13F3-49, B13F3-87, B13F3-97)

|  |                                                                                                                                                            |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | 1 Using the manufacturer approved diagnostic system, clear the DTCs                                                                                        |
|  | 2 Start the engine                                                                                                                                         |
|  | 3 Turn A/C on and set to Hi/Hi                                                                                                                             |
|  | 4 Run engine to operating temperature to allow ATCM to provide hot air                                                                                     |
|  | 5 Toggle A/C settings from LO/LO to HI/HI                                                                                                                  |
|  | <p>When in LO/LO is air from right auxiliary vent cold (&lt;10'c) and when in HI/HI is air from right auxiliary vent hot (&gt;50'c)?</p> <p><b>Yes</b></p> |

Check the DTC's - has the DTC recurred? **A)** If the DTC has recurred go back to step 1 and repeat - if the DTC consistently recurring, check for obstruction/mechanical linkage problem/damage to recirculation door, if fault found repair. Clear the DTCs and retest. **B)** If the DTC has not recurred - no fault found

**No**

Check for obstruction/mechanical linkage problem/damage to recirculation door, if fault found repair. Clear the DTCs and retest. Check wiring - does right auxiliary climate control temperature blend motor have power, ground and LIN connection to ECU? **A)** If wiring or LIN fault found - repair wiring. Clear the DTCs and retest. **B)** If no fault found, and there is sufficient power, ground and LIN connection is good, install a new right auxiliary climate control temperature blend motor. Clear the DTCs and retest

**E9: FRONT COOL AIR BYPASS MOTOR OPERATION (B13EE-49, B13EE-87, B13EE-97)**

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>1</b> Using the manufacturer approved diagnostic system, clear the DTCs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|  | <b>2</b> Start the engine                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|  | <b>3</b> Turn A/C on and set to Hi/Hi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|  | <b>4</b> Run engine to operating temperature to allow ATCM to provide hot air                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|  | <b>5</b> Toggle A/C settings from LO/LO to HI/HI                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|  | <p>When in LO/LO is air from front vents cold (&lt;10'c) and when in HI/HI is air from front vents hot (&gt;50'c)?</p> <p><b>Yes</b></p> <p>Check the DTC's - has the DTC recurred? <b>A)</b> If the DTC has recurred go back to step 1 and repeat - if the DTC consistently recurring, check for obstruction/mechanical linkage problem/damage to recirculation door, if fault found repair. Clear the DTCs and retest. <b>B)</b> If the DTC has not recurred - no fault found</p> <p><b>No</b></p> <p>Check for obstruction/mechanical linkage problem/damage to recirculation door, if fault found repair. Clear the DTCs and retest. Check wiring - does cool air bypass motor have power, ground and LIN connection to ECU? <b>A)</b> If wiring or LIN fault found - repair wiring. Clear the DTCs and retest. <b>B)</b> If no fault found, and there is sufficient power, ground and LIN connection is good, install a new cool air bypass motor. Clear the DTCs and retest</p> |

**E10: DEMIST DISTRIBUTION MOTOR OPERATION (B1085-49, B1085-87, B1085-97)**

|  |                                                                            |
|--|----------------------------------------------------------------------------|
|  | <b>1</b> Using the manufacturer approved diagnostic system, clear the DTCs |
|  | <b>2</b> Start the engine                                                  |
|  | <b>3</b> Turn A/C on and set to Hi/Hi                                      |
|  |                                                                            |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p><b>4</b> Toggle distribution mode between screen only and face only</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|  | <p>Does air come out of the defrost vents when in screen only mode and does no air come out of the defrost vents when in face only mode?</p> <p><b>Yes</b></p> <p>Check the DTC's - has the DTC recurred? <b>A)</b> If the DTC has recurred go back to step 1 and repeat - if the DTC consistently recurring, check for obstruction/mechanical linkage problem/damage to recirculation door, if fault found repair. Clear the DTCs and retest. <b>B)</b> If the DTC has not recurred - no fault found</p> <p><b>No</b></p> <p>Check for obstruction/mechanical linkage problem/damage to recirculation door, if fault found repair. Clear the DTCs and retest. Check wiring - does demist distribution motor have power, ground and LIN connection to ECU? <b>A)</b> If wiring or LIN fault found - repair wiring. Clear the DTCs and retest. <b>B)</b> If no fault found, and there is sufficient power, ground and LIN connection is good, install a new demist distribution motor. Clear the DTCs and retest</p> |

**E11: LEFT FRONT FACE/FEET DISTRIBUTION MOTOR OPERATION (B13EF-49, B13EF-87, B13EF-97)**

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p><b>1</b> Using the manufacturer approved diagnostic system, clear the DTCs</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|  | <p><b>2</b> Start the engine</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|  | <p><b>3</b> Turn A/C on and set to Hi/Hi</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|  | <p><b>4</b> Toggle left front distribution mode between face and foot</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|  | <p>Does air come out of the left face vents when in face mode and does air come out of the left foot vents when in foot mode?</p> <p><b>Yes</b></p> <p>Check the DTC's - has the DTC recurred? <b>A)</b> If the DTC has recurred go back to step 1 and repeat - if the DTC consistently recurring, check for obstruction/mechanical linkage problem/damage to recirculation door, if fault found repair. Clear the DTCs and retest. <b>B)</b> If the DTC has not recurred - no fault found</p> <p><b>No</b></p> <p>Check for obstruction/mechanical linkage problem/damage to recirculation door, if fault found repair. Clear the DTCs and retest. Check wiring - does left front face/feet distribution motor have power, ground and LIN connection to ECU? <b>A)</b> If wiring or LIN fault found - repair wiring. Clear the DTCs and retest. <b>B)</b> If no fault found, and there is sufficient power, ground and LIN connection is good, install a new left front face/feet distribution motor. Clear the DTCs and retest</p> |

**E12: RIGHT FRONT FACE/FEET DISTRIBUTION MOTOR OPERATION (B13F0-49, B13F0-87, B13F0-97)**

|  |                                                                                   |
|--|-----------------------------------------------------------------------------------|
|  | <p><b>1</b> Using the manufacturer approved diagnostic system, clear the DTCs</p> |
|  | <p><b>2</b> Start the engine</p>                                                  |
|  | <p><b>3</b> Turn A/C on and set to Hi/Hi</p>                                      |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>4</b> Toggle right front distribution mode between face and foot                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|  | <p>Does air come out of the right face vents when in face mode and does air come out of the right foot vents when in foot mode?</p> <p><b>Yes</b></p> <p>Check the DTC's - has the DTC recurred? <b>A)</b> If the DTC has recurred go back to step 1 and repeat - if the DTC consistently recurring, check for obstruction/mechanical linkage problem/damage to recirculation door, if fault found repair. Clear the DTCs and retest. <b>B)</b> If the DTC has not recurred - no fault found</p> <p><b>No</b></p> <p>Check for obstruction/mechanical linkage problem/damage to recirculation door, if fault found repair. Clear the DTCs and retest. Check wiring - does right front face/feet distribution motor have power, ground and LIN connection to ECU? <b>A)</b> If wiring or LIN fault found - repair wiring. Clear the DTCs and retest. <b>B)</b> If no fault found, and there is sufficient power, ground and LIN connection is good, install a new right front face/feet distribution motor. Clear the DTCs and retest</p> |

### E13: REAR FACE/FEET DISTRIBUTION MOTOR OPERATION (B1B7D-49, B1B7D-87, B1B7D-97)

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>1</b> Using the manufacturer approved diagnostic system, clear the DTCs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|  | <b>2</b> Start the engine                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|  | <b>3</b> Turn rear A/C on and set to Hi/Hi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|  | <b>4</b> Toggle rear distribution mode between face and foot                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|  | <p>Does air come out of the rear face vents when in face mode and does air come out of the rear foot vents when in foot mode?</p> <p><b>Yes</b></p> <p>Check the DTC's - has the DTC recurred? <b>A)</b> If the DTC has recurred go back to step 1 and repeat - if the DTC consistently recurring, check for obstruction/mechanical linkage problem/damage to recirculation door, if fault found repair. Clear the DTCs and retest. <b>B)</b> If the DTC has not recurred - no fault found</p> <p><b>No</b></p> <p>Check for obstruction/mechanical linkage problem/damage to recirculation door, if fault found repair. Clear the DTCs and retest. Check wiring - does rear face/feet distribution motor have power, ground and LIN connection to ECU? <b>A)</b> If wiring or LIN fault found - repair wiring. Clear the DTCs and retest. <b>B)</b> If no fault found, and there is sufficient power, ground and LIN connection is good, install a new rear face/feet distribution motor. Clear the DTCs and retest</p> |

### E14: FRONT FACE/FEET DISTRIBUTION MOTOR OPERATION (B1434-49, B1434-87, B1434-97)

|  |                                                                            |
|--|----------------------------------------------------------------------------|
|  | <b>1</b> Using the manufacturer approved diagnostic system, clear the DTCs |
|  | <b>2</b> Start the engine                                                  |
|  |                                                                            |

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|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>3</b> Turn A/C on and set to Hi/Hi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|  | <b>4</b> Toggle front distribution mode between face and foot                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|  | <p>Does air come out of the front face vents when in face mode and does air come out of the front foot vents when in foot mode?</p> <p><b>Yes</b></p> <p>Check the DTC's - has the DTC recurred? <b>A)</b> If the DTC has recurred go back to step 1 and repeat - if the DTC consistently recurring, check for obstruction/mechanical linkage problem/damage to recirculation door, if fault found repair. Clear the DTCs and retest. <b>B)</b> If the DTC has not recurred - no fault found</p> <p><b>No</b></p> <p>Check for obstruction/mechanical linkage problem/damage to recirculation door, if fault found repair. Clear the DTCs and retest. Check wiring - does front face/feet distribution motor have power, ground and LIN connection to ECU? <b>A)</b> If wiring or LIN fault found - repair wiring. Clear the DTCs and retest. <b>B)</b> If no fault found, and there is sufficient power, ground and LIN connection is good, install a new front face/feet distribution motor. Clear the DTCs and retest</p> |

#### E15: AUXILIARY CLIMATE CONTROL FACE/FEET DISTRIBUTION MOTOR OPERATION (B13F1-49, B13F1-87, B13F1-97)

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>1</b> Using the manufacturer approved diagnostic system, clear the DTCs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|  | <b>2</b> Start the engine                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|  | <b>3</b> Turn A/C on and set to Hi/Hi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|  | <b>4</b> Toggle rear climate distribution mode between face and foot                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|  | <p>Does air come out of the face vents when in face mode and does air come out of the foot vents when in foot mode?</p> <p><b>Yes</b></p> <p>Check the DTC's - has the DTC recurred? <b>A)</b> If the DTC has recurred go back to step 1 and repeat - if the DTC consistently recurring, check for obstruction/mechanical linkage problem/damage to recirculation door, if fault found repair. Clear the DTCs and retest. <b>B)</b> If the DTC has not recurred - no fault found</p> <p><b>No</b></p> <p>Check for obstruction/mechanical linkage problem/damage to recirculation door, if fault found repair. Clear the DTCs and retest. Check wiring - does auxiliary climate control face/feet distribution motor have power, ground and LIN connection to ECU? <b>A)</b> If wiring or LIN fault found - repair wiring. Clear the DTCs and retest. <b>B)</b> If no fault found, and there is sufficient power, ground and LIN connection is good, install a new auxiliary climate control face/feet distribution motor. Clear the DTCs and retest</p> |

For a list of Diagnostic Trouble Codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00.

REFER to: [Diagnostic Trouble Code Index - DTC: Automatic Temperature Control Module \(ATCM\) - Hybrid Electric Vehicle](#) (100-00 General Information, Description and Operation).